

UNC-CH STOR 538

NBA PREDICTION PROJECT

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April 9, 2024

Data Information

In order to build our dataset, we first used an API built in Python to pull Advanced Box Scores data from nba.com. We gathered data from all games occurring between the beginning of the 2019-2020 regular season up until April 7, 2024. From the API, we collected the following game metrics for each team: Team Name, Team Abbreviation, Game ID (*GAMEID*), Game Date (*GAME_DATE*), Matchup, Win/Loss (*WL*), Minutes (*MIN*), Points (*PTS*), Field Goals Made (*FGM*), Field Goals Attempted (*FGA*), Field Goal Percentage (*FG_PCT*), Three-Point Field Goals Made (*FG3M*), Three-Point Field Goals Attempted (*FG3A*), Three-Point Field Goal Percentage (*FG3_PCT*), Free Throws Made (*FTM*), Free Throws Attempted (*FTA*), Free Throw Percentage (*FT_PCT*), Offensive Rebounds (*OREB*), Defensive Rebounds (*DREB*), Rebounds (*REB*), Assists (*AST*), Steals (*STL*), Blocks (*BLK*), Turnovers (*TO*), Personal Fouls (*PF*), and Plus/Minus (*PLUS_MINUS*). This dataset had two rows for one game, so our next step was to convert the dataset to have one row per game with statistics for both the home and away teams. To do this, we first created a “Home” and “Away” column for each row using the Matchup variable that contained the team name of the home and away teams for each game. We assigned rows that contained ‘vs.’ in the Matchup column as away teams and rows that contained ‘@’ in the Matchup column as home teams. We then split the dataset into two separate dataframes, one containing the data for the home teams and the other containing the data for away teams. Next, we renamed the variables to have ‘H’ in front for the home dataset and ‘A’ in front for the away dataset in order to differentiate the variables after we combine the two datasets. We then did a full join of the home and away datasets using *GAMEID* and *GAME_DATE* variables as the join criteria in order to combine all home and away data from each game into one row. Next, we mutated this combined dataset to include *SPREAD*, *TOTAL*, and Total *OREB* variables. *SPREAD* was calculated by Home Points (*HPTS*) - Away Points (*APTS*), *TOTAL* was calculated by *HPTS* + *APTS*, and Total *OREB* was calculated by Home *OREB* (*HOREB*) + Away *OREB* (*AOREB*). Finally, we added a column called *season_type* that indicated whether each game occurred during Preseason, Regular Season, or Postseason. Any games that included *NA* values for their box score data were not included in our analysis.

Only the regular season games were included in the final dataset. All preseason and summer league games were excluded due to the vastly different nature of these games and a general lack of incentive to win compared to the regular season games. All playoff games were also excluded since these games tend to be much different from regular season NBA games, with the 7 game series format potentially leading to confounding effects on the game observations. The *GAME_DATE* variable for each game was converted to a date-type to allow for easier data manipulation with identifying start and end dates of each regular season in the data set.

Using the metrics in our original dataset we got from the API, we created an assist to turnover ratio, shooting efficiency metric (points per attempts), effective field goal percentage (*eFG%*), assist percentage (assists per field goals made), and two different types of rebound percentage. One of the rebound percentages, *hrebounnd_ratio* or *arebounnd_ratio*, is the percentage of rebounds the team got considering all rebounds gotten in the game. The other rebound percentage, *hrebounnd_FGM* or *arebounnd_FGM*, is the team’s rebounds they got out of all field goals missed, because each missed field goal would result in a rebound. Like in our original dataset, these variables were named with ‘H’ in front

to indicate that it is the home team, and with an 'A' in the front to indicate that it is the away team. The following list contains the calculations for each new variable in the dataset:

$$\begin{aligned}
 ast_{tov} &= \frac{AST}{TOV} \\
 shooting_efficiency &= \frac{PTS}{FGA} \\
 eFG\% &= \frac{FGM + (0.5 * 3PM)}{FGA} \\
 assist_percentage &= \frac{AST}{FGM} * 100 \\
 rebound_ratio &= \frac{OREB + DREB}{HOREB + HDREB + AOREB + ADREB} \\
 rebound_FGM &= \frac{OREB + DREB}{(HFGA + AFGA) - (HFGM - AFGM)}
 \end{aligned}$$

We chose these ratios for different reasons. Each ratio was made based on our evaluation of the data, and we determined that each can likely provide helpful insights to prediction for game results. Rebounds give a team the possession of the ball, opening up scoring opportunities. Knowing the rebound ratio is important because it reveals how often a team took advantage of opportunities to get a rebound. The assist percentage and shooting efficiency were two other created metrics. The assist percentage shows how often a field goal made was the result of an assist, which can show how well the team moves the ball around to create scoring opportunities. The more assists there are, the more the team is working together, and this should result in a higher score. The shooting efficiency metric measures how many points a team scores out of field goal attempts, and the higher this is, the more consistently they scored. This also would be higher if a team scored a lot of three-pointers or converted a large amount of free throws. Having a higher ratio of points to field goal attempts would be predictive of a higher score because the team was more efficient offensively, thus wasting less possessions of the ball. *eFG%* for each team was calculated and added to each game in the data set. This variable was created by Dean Oliver, and it is a way to more effectively weight the 3-point shot when considering a team's field goal percentage. This variable was added to gain a better assessment of a team's offensive abilities and to allow for later removal of variables such as *FGM* and *3PM* which cannot be included into any predictive models of points scored since they are the variables which directly contribute to a team's final point total.

To be able to turn the game statistics we scraped into predictions for the future, we had to find a way to calculate statistics for games we did not yet have information for. We knew that for the final predictions we would only be provided with the name of the home team and away team, so we could not use models that took specific metrics from an individual game (i.e. home rebounds, away assists, etc.) to predict *SPREAD* and *OREB* on that day. We decided that the best way to use the data we had from previous games to predict future games was to calculate average statistics from each team in various categories (points, rebounds, assists, etc. per game). To calculate team averages, the data was split into home and away performances. We then grouped the data by team to calculate the averages for when each team was home versus when they were away. After, we recombined the home and away data to create the average data for the matchups during each season. After combining the data, each line contained the average box score data for the home team and the away team in each game. This new average data for each team allowed us to create informed models based on past performance when looking at future games with no data already available. In our models, the box score data beginning with 'H' or 'A' represents the

average data for the home and away team in a given season (i.e. *HFG_PCT* represents the average field goal percentage that the home team in a given matchup had during the season).

In order to add more data to use in our models, we employed the same API that was used to gather advanced box score data to pull team advanced statistics on each team from nba.com. From this API, we collected the following season metrics for each team: Games Played (*GP*), Wins, Losses, Total Minutes Played (*MIN*), Offensive Rating (*OFF_RATING*), Defensive Rating (*DEF_RATING*), Net Rating (*NET_RATING*), Offensive Rebound Percentage (*OREB_PCT*), Defensive Rebound Percentage (*DREB_PCT*), Rebound Percentage (*REB_PCT*), and Average Number of Possessions per Game (*PACE*). Offensive rating represents how many points a given team typically scores over 100 possessions, while defensive rating represents how many points a given team's opponents typically score over 100 possessions. We then combined this data with our cleaned advanced box score dataset using a left join. We first joined based on the home team and renamed the added variables to begin with 'H.' We then repeated this step using the away team, renaming the added variables to begin with 'A.' Using these variables, we created two new measures called Home Efficiency and Away Efficiency in order to factor in the opponent's defensive ability to each team's offensive rating. Home Efficiency and Away Efficiency were calculated using the following formulas:

$$\begin{aligned} \text{Home_Efficiency} &= \frac{H_OFF_RATING + A_DEF_RATING}{2} \\ \text{Away_Efficiency} &= \frac{A_OFF_RATING + H_DEF_RATING}{2} \end{aligned}$$

We also created a variable called Average Possessions, calculated by the following:

$$\text{Average_Possessions} = \frac{H_PACE + A_PACE}{2}$$

This measured the average number of possessions in specific matchups. Since *SPREAD*, *TOTAL*, and offensive rebounds are highly related to a team's offensive ability, we believed that including these advanced statistics would improve our ability to predict these variables. We believed net rating, which is the difference between offensive and defensive rating, would be useful in predicting *SPREAD* since it shows, on average, the difference in points scored between a given team and their opponents. We also believed that creating the home and away efficiency variables that compared how specific teams played one another would help us predict both *SPREAD* and *TOTAL* more accurately for specific matchups. Finally, we thought that using variables such as *PACE* and average number of possessions would be useful because teams that play more possessions would have more opportunities to score more points or get more offensive rebounds.

We then used the same API to pull lineup data for each team from nba.com. From this API, we collected the following average seasonal metrics for every lineup that each team used: Team Name, Players in each Lineup, Games Played (*GP*), Wins, Losses, Minutes (*MIN*), Points (*PTS*), Field Goals Made (*FGM*), Field Goals Attempted (*FGA*), Field Goal Percentage (*FG_PCT*), Three-Point Field Goals Made (*FG3M*), Three-Point Field Goals Attempted (*FG3A*), Three-Point Field Goal Percentage (*FG3_PCT*), Free Throws Made (*FTM*), Free Throws Attempted (*FTA*), Free Throw Percentage (*FT_PCT*), Offensive Rebounds (*OREB*), Defensive Rebounds (*DREB*), Rebounds (*REB*), Assists (*AST*),

Steals (*STL*), Blocks (*BLK*), Turnovers (*TO*), Personal Fouls (*PF*), and Plus/Minus (*PLUS_MINUS*). We then created a variable called Average Game Time (*AvgGameTime*), which was calculated by $\text{MIN}/48$, in order to determine the average percentage of game time each lineup plays during a typical 48 minute NBA game. We also created a variable called Average Season Time (*AvgSeasonTime*), which was calculated by $\text{GP}/82$, in order to determine the percentage of games in which each lineup played in a 82 game NBA regular season. We then used these variables to create the following:

$$\begin{aligned}
 \text{Average Game Plus/Minus (AvgGamePM)} &= \text{PLUS_MINUS} * \text{AvgGameTime} \\
 \text{Average Season Plus/Minus (AvgSeasonPM)} &= \text{AvgGamePM} * \text{GP} \\
 \text{Average Game OREB (AvgGameOREB)} &= \text{OREB} * \text{AvgGameTime} \\
 \text{Average Season OREB (AvgSeasonOREB)} &= \text{AvgGameOREB} * \text{GP} \\
 \text{Average Game DREB (AvgGameDREB)} &= \text{DREB} * \text{AvgGameTime} \\
 \text{Average Season DREB (AvgSeasonDREB)} &= \text{AvgGameDREB} * \text{GP} \\
 \text{Average Game REB (AvgGameREB)} &= \text{REB} * \text{AvgGameTime} \\
 \text{Average Season REB (AvgSeasonREB)} &= \text{AvgGameREB} * \text{GP}
 \end{aligned}$$

Most of the lineup data from nba.com is given in terms of full games, so we created the above variables in order to better reflect how specific lineups contributed based on the time they are actually playing in the game and the number of games in the season they actually play in. We then grouped the lineup data by team in order to calculate the average values of the variables for each team. Next, we combined the variables that we created using the lineup data with our dataset including the advanced box score and advanced statistics using a left join. We first joined based on the home team and renamed the added variables to begin with 'H.' We then repeated this step using the away team, renaming the added variables to begin with 'A' in order to differentiate them from the home team variables. Since the chemistry of lineups and the ability of players to perform well together on the court is a large factor in how well a team plays, we believed that lineup data would be beneficial to include in our models. Plus/Minus indicates the point differential between a certain lineup and their opponents. We believed that using plus/minus combined with the fraction of time specific lineups played would be beneficial in determining the overall *SPREAD* of games. We also believed that factoring the rebounding ability of specific lineups into our models would help better predict total *OREB*. Some teams could choose to play a lineup of taller players more often versus smaller shooters, which could result in more *OREB*.

When creating our models, we used the 2023 regular season data as our training data, which occurred from October 18, 2022 to April 3rd, 2023. We then tested these models on the data from the last week of the regular season during the 2022-23 season, which occurred from April 4th to April 9th of 2023. Some models were also tested through leave-one-out cross validation on each game of the 2022-2023 season. We initially created our models using data from each season since the 2019-2020 regular season but found the predictions to be more accurate when only using the current season's data. This could be the case due to roster changes or coaching decisions that differ between seasons.

Spread Methodology

To identify the best model to predict the *SPREAD* of NBA games, we first used the season average points per game during the 2022-2023 season and assigned this value to each team, regardless of the point in the season. We then found the *SPREAD* of each game in the 2022-2023 season using these averaged values, and calculated the MAE. Next, we followed the same process to predict games in the 2023 season, but instead used averages from all regular season games from 2019-2023 for each team. We again calculated the MAE for these predictions.

Next we created a multiple linear regression model containing all relevant variables in the NBA dataset to predict the points scored by each team in a game. To do this, we had to elongate the data set to contain two observations for each game. The first of the two observations contained data for the home team, identifiable as *TEAM*, and the away team as *Opponent*. The second observation was the opposite. This allowed us to use the following parameters to predict points for a given team in a game (*Team_PTS*):

Team_FGM, Team_FGA, Team_FG_PCT, Team_FG3M, Team_FG3A, Team_FG3_PCT, Team_FTM, FTA, Team_FT_PCT, Team_OREB, Team_DREB, Team_AST, Team_STL, Team_BLK, Team_TOV, Team_PF

Since this model contained the number of 2-point and 3-point field goals made, as well as free throws made, the model had an R-squared value of 1. This was expected due to the nature of the model containing these variables. Additionally, the VIF value of this model showed several instances of multicollinearity. To combat this, a new linear regression model was fitted after the following variables were removed: *Team_FGM, Team_FGA, Team_FG3M, Team_FG3A, Team_FTM, Team_FTA*. This model showed to be effective at predicting the point total of NBA teams. This model fit the assumptions of no multicollinearity, as no VIF was higher than 2.

We then fit a third linear regression model containing all of the same predictor variables but replacing *Team_FG_PCT* and *Team_FG3_PCT* with the eFG% of the team (*Team_EFG*). This model showed to be an improvement on the previous model. This model also fit the assumptions of no multicollinearity.

To perform one final check on the combination of predictor variables, each combination of predictor variables from the 9 variables used in the previous model were used in a linear regression model. The AIC and R-squared values were then compared between each of the 512 possible models without interactions, and the model containing all 9 variables had the both lowest AIC and highest R-squared value. The table containing the top 5 models and their respective AIC and R-squared values are shown below:

Formula	Adjusted R ²	AIC
<i>Team_PTS</i> ~ <i>Team_EFG</i> + <i>Team_FT_PCT</i> + <i>Team_OREB</i> + <i>Team_DREB</i> + <i>Team_AST</i> + <i>Team_STL</i> + <i>Team_BLK</i> + <i>Team_TOV</i> + <i>Team_PF</i>	0.777	80349.79
<i>Team_PTS</i> ~ <i>Team_EFG</i> + <i>Team_FT_PCT</i> + <i>Team_OREB</i> + <i>Team_DREB</i> + <i>Team_AST</i> + <i>Team_STL</i> + <i>Team_TOV</i> + <i>Team_PF</i>	0.777	80354.47
<i>Team_PTS</i> ~ <i>Team_EFG</i> + <i>Team_FT_PCT</i> + <i>Team_OREB</i> + <i>Team_DREB</i> + <i>Team_STL</i> + <i>Team_BLK</i> + <i>Team_TOV</i> + <i>Team_PF</i>	0.770	80751.56
<i>Team_PTS</i> ~ <i>Team_EFG</i> + <i>Team_FT_PCT</i> + <i>Team_OREB</i> + <i>Team_DREB</i> + <i>Team_STL</i> + <i>Team_TOV</i> + <i>Team_PF</i>	0.769	80760.83
<i>Team_PTS</i> ~ <i>Team_EFG</i> + <i>Team_FT_PCT</i> + <i>Team_OREB</i> + <i>Team_DREB</i> + <i>Team_AST</i> + <i>Team_BLK</i> + <i>Team_TOV</i> + <i>Team_PF</i>	0.766	80941.26

The table above shows that the best combination of predictor variables was in fact the aforementioned model containing all 9 predictor variables.

Once the best linear regression model for predicting team points was created, the model was then used to predict the *SPREAD* of each game in the 2022-2023 season using the season averages for each team using leave-one-out cross validation. The total points scored for each team was calculated as the mean prediction of *Team_PTS* of the and the mean points allowed of the opposing team (*Opp_Points*). The effectiveness of the predictions was then measured using MAE.

After factoring in the opponent using the linear model to make predictions, we then factored in the home team advantage as well. To do this, another set of averages were calculated for the 2023 season for each team both as the home team and as the away team. This provided 60 different “teams” to work with, where each team was represented by averages separately as the home team and the away team. The method to predict the *SPREAD* of each game in the 2022-2023 regular season was then determined using the same method as before with the linear regression model and the average of the team points and the opponent points allowed.

Lastly, the season averages for all regular season games in the 2019-2023 seasons were used in the same method to predict the *SPREAD* of 2022-2023 regular season games. The accuracy of the predictions was again measured using MAE. The resulting MAE for each model can be seen in the table at the end of this section.

An additional predictor we attempted to include was data from different lineups that have played on each team during the season. The lineup data included insights like pace and player impact estimate,

each of which are advanced metrics. Our goal was to determine if pace, player impact estimate, or games and minutes played by a lineup could strengthen our prediction of home and away team scores, again basing it off of 2023 data so we could check our predictions on games that have already happened. After scraping the data off of the NBA's website, we subsetting that data to include only the most common lineup for each team, based on minutes played. Using pace and player impact estimate as predictors in this model, each made little impact on the prediction model. An additional way we used the pace and player impact estimate from this model was to average it for all lineups by team, using the 'group_by' and 'summarize_all' functions. Applying these to our linear model for prediction, there was also no big impact. The data from the lineups were not significant in helping our prediction model. This could be because there is a lot of variability, even when using the same lineup. There are differences in how they play and interact with each game.

The ratios we added in the data information section were additionally tested as predictors for the last week of the 2023 season. Using 2023 season averages to find the ratios and only having these ratios as predictors, the best home and away score linear regression models ended up with a MAE for *SPREAD* of 14.955. The *SPREAD* model using ratios is listed below.

$$\text{Spread_Model} = \text{lm}(\text{HPTS} \sim \text{Home} + \text{hast_tov} + \text{hshooting_efficiency} + \text{hassist_percentage} + \text{hreboun_ratio} + \text{HEFG} + \text{hreboun_FGM}) - (\text{APTS} \sim \text{Away} + \text{aast_tov} + \text{ashooting_efficiency} + \text{aassist_percentage} + \text{areboun_ratio} + \text{AEFG} + \text{areboun_FGM})$$

$$\text{Spread_Model} = \text{lm}(\text{HPTS} \sim \text{Home} + \text{hast_tov} + \text{hshooting_efficiency} + \text{hassist_percentage} + \text{hreboun_ratio} + \text{HEFG} + \text{hreboun_FGM}) - (\text{APTS} \sim \text{Away} + \text{aast_tov} + \text{ashooting_efficiency} + \text{aassist_percentage} + \text{areboun_ratio} + \text{AEFG} + \text{areboun_FGM})$$

In addition to the linear models explored, we also investigated the use of Random Forest models, a popular machine learning method that constructs multiple decision trees during training and outputs the average prediction of the individual trees. Specifically, we used 500 decision trees. Despite its ability to capture intricate patterns in the data, the Random Forest model yielded a MAE of 11.971 when trained on 2023 data only, indicating slightly less accuracy compared to the linear models. Despite its ability to capture intricate patterns in the data, the Random Forest model may not have performed as well as the linear models due to potential overfitting and the complexity of the model. Additionally, fine-tuning the model's hyperparameters could be challenging and may require further optimization to enhance its predictive accuracy.

Based on this process, it was reasonable to conclude from MAE that the model training on 2023 data that considered the opponent and if the team was home or away best predicted *SPREAD*. The table below summarizes the results we got from each model.

Model	Season Data	MAE for 2023 SPREAD Prediction
Season averages only	2023	10.824
Season averages only	2019-2023	11.566
Linear model with opponent consideration	2023	10.480
Linear model with opponent and home/away consideration	2023	10.211
Linear model with opponent and home/away consideration	2019-2023	11.280
Random Forest model	2023	11.971
Season averages, ratios as predictors	2023	14.955

As seen in the above table, the best predictor model for *SPREAD* was the multiple linear regression model which contained information regarding both opponent and home/away status. The final linear regression model to predict *Team_PTS* with respective coefficients is below:

$$Team_PTS = 1.5(Team_EFG) + 0.198(Team_FT_PCT) + 1.05(Team_OREB) + 0.301(Team_DREB) + 0.270(Team_AST) + 0.467(Team_STL) + 0.0564(Team_BLK) - 0.724(Team_TOV) + 0.480(Team_PF)$$

This can be interpreted as for every one unit increase in a given statistic, the predicted point total of that team increases by the amount of the respective coefficient. Since eFG% is written in the dataset as a percentage rather than a ratio (i.e. 56%, not 0.56), this model indicates that a 1% increase in eFG% for a team will increase their predicted total by 1.53 points.

Testing our additional models reassured that our best prediction model is the one listed below. This model was applied to the 2024 season to get our final predictions for *SPREAD*. It is important to note that for the following equations, *H_Team* refers to the home team, and *A_Team* refers to the away team. Additionally, *Team* and *Opponent* are relative to the team in which we are predicting. For example, in predicting the points scored of the home team – *H_PTS_Scored* – we predict *Team_PTS* using the home team's data, and we predict *Opp_PTS* using the mean points allowed by the away team.

$$H_PTS_Scored = lm(Team_PTS \sim 1.53(Team_EFG) + 0.198(Team_FT_PCT) + 1.05(Team_OREB) + 0.301(Team_DREB) + 0.270(Team_AST) + 0.467(Team_STL) + 0.0564(Team_BLK) - 0.724(Team_TOV) + 0.480(Team_PF))$$

$$H_PTS_Allowed = lm(Opp_PTS \sim 1.53(Opp_EFG) + 0.198(Opp_FT_PCT) + 1.05(Opp_OREB) + 0.301(Opp_DREB) + 0.270(Opp_AST) + 0.467(Opp_STL) + 0.0564(Opp_BLK) - 0.724(Opp_TOV) + 0.480(Opp_PF))$$

$$A_PTS_Scored = lm(Team_PTS \sim 1.53(Team_EFG) + 0.198(Team_FT_PCT) + 1.05(Team_OREB) + 0.301(Team_DREB) + 0.270(Team_AST) + 0.467(Team_STL) + 0.0564(Team_BLK) - 0.724(Team_TOV) + 0.480(Team_PF))$$

$$A_PTS_Allowed = lm(Opp_PTS \sim 1.53(Opp_EFG) + 0.198(Opp_FT_PCT) + 1.05(Opp_OREB) + 0.301(Opp_DREB) + 0.270(Opp_AST) + 0.467(Opp_STL) + 0.0564(Opp_BLK) - 0.724(Opp_TOV) + 0.480(Opp_PF))$$

$$H_PTS = \frac{H_PTS_Scored + A_PTS_Allowed}{2}$$

$$A_PTS = \frac{A_PTS_Scored + H_PTS_Allowed}{2}$$

$$Spread_Model = H_PTS - A_PTS$$

Total Methodology

To predict the *TOTAL* variable, we used the same methodology as we did for the *SPREAD* variable. Our plan to predict the *SPREAD* was to work based on the home team score and the away team score, and our plan to predict *TOTAL* was the same. The only difference was the calculation of the final prediction. The variable was calculated by adding the predicted home score and the predicted away score together. The entire process was the same until that point.

Like in the *SPREAD* variable prediction, we tested out many combinations of predictors, including the ratio variables we created. In predicting the *TOTAL* of NBA games, the focus extends beyond the simple comparison of team performance in *SPREAD* prediction. *TOTAL* prediction requires a comprehensive understanding of offensive strategies, defensive capabilities, and overall team dynamics. The model's ability to forecast *TOTAL* relies heavily on its ability to capture the combined impact of various factors influencing both the home and away teams' offensive output. Unlike *SPREAD* prediction, which primarily considers the performance differential between teams, *TOTAL* prediction requires focus on how that team usually performs based on score.

We once again attempted to use Random Forest models to see how its accuracy would compare to the more traditional linear models. Despite having a comparable MAE, the Random Forest models encountered challenges with overfitting and excessive complexity, ultimately falling short of the straightforward accuracy achieved by the linear model.

The optimal model to predict *TOTAL* ultimately was much more simple than prediction of the *SPREAD*. When predicting each regular season game in the 2022-2023 season, the data that most accurately predicted the score used season points per game averages from only the 2022-2023 season. Only using the most recent season likely results in the best outcome because it better represents current trends in the game. Overall offensive performance for teams changes from year to year. The data set that best predicts *TOTAL* is likely different from the one for *SPREAD* because of the differences in what each model focuses on. Since *SPREAD* is focused on the difference in what two teams score against each other,

accounting for opponent and home or away is necessary to capture the difference in how teams perform against each other. Predicting *TOTAL* requires more focus on a team's typical offensive performance, which is best encapsulated by their season average regardless of opponent. The table below summarizes the findings for *TOTAL*:

Model	MAE for 2023-24 <i>TOTAL</i>
Season averages with 2023 data only	14.748
Season averages with 2019-2023 data only	15.682
Linear model with opponent consideration	15.243
Linear model with opponent and home/away consideration, 2023 data	15.131
Random Forest model with 2023 data only	15.815
Linear model with opponent and home/away consideration, 2019-2023 data	15.929
Season averages with 2023 data only, ratios as predictors	20.904

We applied our model that best predicted *TOTAL* for the 2023 regular season games to the 2024 season. Since the model we found to be the most accurate only added the team's average scores together, the model that we used to make our prediction is as follows:

$$TOTAL = HPTS + APTS$$

The *TOTAL* prediction variable was calculated by adding the home team score average and the away team score average, without using any other metrics or seasons of data. The averages were updated for each team as of April 7, 2024, and can be seen in the table below:

Team	Points Per Game
Atlanta Hawks	119.7
Boston Celtics	120.8
Brooklyn Nets	112.5
Charlotte Hornets	107.5
Chicago Bulls	111.3

Cleveland Cavaliers	112.4
Dallas Mavericks	118.7
Denver Nuggets	114.9
Detroit Pistons	110.3
Golden State Warriors	117.5
Houston Rockets	113.6
Indiana Pacers	123.7
Los Angeles Clippers	116.2
Los Angeles Lakers	116.8
Memphis Grizzlies	106.2
Miami Heat	110.9
Milwaukee Bucks	121.4
Minnesota Timberwolves	112.9
New Orleans Pelicans	115.3
New York Knicks	113.2
Oklahoma City Thunder	120.3
Orlando Magic	111.2
Philadelphia 76ers	116.2
Phoenix Suns	116.3
Portland Trail Blazers	107.6
Sacramento Kings	117.0
San Antonio Spurs	111.9
Toronto Raptors	112.7
Utah Jazz	115.2
Washington Wizards	114.3

Although this model is not very complex, it performed best given the NBA game data compared to all other models we looked at. One explanation for *TOTAL* being difficult to predict could be the drastic

increase in scoring in the NBA over the past decade. This could make other predictive models less effective as the increase in points per season was not accounted for in any of our models, thus causing *TOTAL* to be more inaccurate than other measures, like *SPREAD*.

Offensive Rebound Methodology

To build our *OREB* models, we first began by using only the data from the average advanced box score variables. The variables coming from the box score data with 'H' in front represent the average value for each team when they play at home, while the variables with 'A' in front do the same for each team when they play away. We created multiple simple models and compared which combination of predictors resulted in the lowest MAE between the actual and predicted *OREB* values. In order to find the best combination of statistically significant predictors, we used stepwise regression. This resulted in the following model:

$$OREBModel1 = lm(OREB \sim HTotalOREB + ATotalOREB + HFG_PCT + AFG_PCT + HFGA + ASpread)$$

AvgHTotalOREB represents the average total number of *OREB* that both teams score when a given team plays at home. *AvgATotalOREB* represents the average total number of *OREB* that both teams score when a given team plays at away. For example, in the 2022-23 when the Brooklyn Nets played at home, the average sum of the *OREB* scored by the Nets and the team they were playing was 18.950. However, when the Brooklyn Nets did not play at home, the average sum of the *OREB* scored by the Nets and the team they were playing was 20.230. When fitting the data to the training set, the model resulted in a MAE of 4.126. We then used this model to predict the *OREB* values from our test set, which resulted in a MAE of 3.635. The full model that used every variable available resulted in a MAE of 4.087 when used on the training set and a MAE of 3.640 when used on the test set. Since the goal of our model is to predict the *OREB* values during the last week of the season, we decided to prioritize the lower MAE value on the test set versus the training set, which is why we considered this model better than the full one. Also, the MAE values are very similar to one another, meaning that the full model likely had numerous predictors that were not statistically significant. We also created numerous other models with different combinations of advanced box score statistics that we thought would be useful in predicting *OREB*, but found that the above model was the most accurate at predicting.

We then created multiple models that also used the data that we collected from the advanced statistics on each team from nba.com. After trying numerous different combinations of predictors and using stepwise regression, we found that the best model that contained this data, in addition to the advanced box score averages, was the following:

$$OREBModel2 = lm(OREB \sim HTotalOREB + ATotalOREB + HFG_PCT + AFG_PCT + HFGA + HTotal + ATotal + ASpread + AOREB_PCT + HOREB_PCT + HPACE + APACE + HomeEff + AwayEff)$$

When fitting the data to the training set, the model resulted in a MAE of 4.196. We then used this model to predict the *OREB* values from our test set, which resulted in a MAE of 3.693. Since both of these MAE values were higher than those from our first model, we found that just adding advanced statistics like offensive rebound percentage or efficiency ratings does not improve models for *OREB*.

Next, we created models using the lineup data, advanced statistics, and advanced box score averages. We performed a stepwise regression to find the best combination of predictors, which resulted in the following model:

$$\text{OREBModel3} = \text{lm}(\text{OREB} \sim \text{HTotalOREB} + \text{ATotalOREB} + \text{HFGA} + \text{HTotal} + \text{ATotal} + \text{ASpread} + \text{HAvgGamePM} + \text{AAvgGamePM})$$

When fitting the data to the training for the 2022-23 set, the model resulted in a MAE of 4.125. We then used this model to predict the *OREB* values from our test set from the same season, which resulted in a MAE of 3.625. These MAE values were very similar to those we found from our first model, so we decided to test both on the data from the 2023-24 season in order to see which performed better. The full model that used every variable we collected had a MAE of 4.082 when applied to the training set and a MAE of 3.883 when applied to the testing set. The predictions for the last week of the season were slightly worse than our predictions from OREBModel1 and OREBModel3, which is why we chose this model over the full one.

When fitting the models to the 2023-24 season, we used data on games from the first day of the 2023-24 regular season, which was October 24, 2023, up until April 7, 2024. Using OREBModel1 with this data resulted in a MAE of 3.832, while using OREBModel3 with this data resulted in a MAE of 3.833. The R-squared value for both of these models was 0.144. Although this is lower than we would like it to be, we were unable to find a combination of predictors from our variables that had a higher R-squared value. Since the models essentially predicted the data just as well as one another, we decided to move forward with the first model since it only requires data from the advanced box score, making it a more simple yet just as effective model.

After deciding on our best linear model, we performed numerous nonlinear transformations in an attempt to improve the model. The residuals for our model had a slight positive skew, so we decided to start with a logarithmic transformation to fix this. Taking the log of the Y variable resulted in a model with an R-squared value of 0.141 and a MAE between actual and predicted *OREB* values of 3.852. These values were similar to those of the original model and the residual plots did not display much improvement, so we decided that this transformation was not helpful. We then used a log transformation on the X variables, which resulted in a model with an R-squared of 0.143 and a MAE of 3.852. Again, these values and the residuals were very similar to those of the original model. When applying a log transformation to both the independent and dependent variables, the resulting model had an R-squared value of 0.016 and a MAE of 4.15. We attempted various other combinations of nonlinear transformations such as square roots, exponentiation and reciprocals, but found none that improved our model.

Another method we considered using when creating models for offensive rebounding was looking at the interactions between multiple different variables. While it can be valuable to use each statistic on

its own, it is often beneficial to use the interaction between more than one variable to make a more accurate model. After looking at simple linear models using solo statistics, we decided to use interactions between some of the variables such as average team assists times average team turnovers, average team steals times average team blocks, etc. When looking at the p-values for the various interactions, it appeared that most of the interactions had very large p-values signaling weak relation with the accuracy of the model; however, the interaction between team assists and team steals had a relatively low p-value. This meant that there was a chance steals and assists could be used together in unison to strengthen our model. We created another model that cleared out the less important interaction effects and focused on the interaction between assists and steals. This resulted in a very high mean absolute error (MAE of 33.756 and R-Squared value of 0.231), which meant the model was not useful. While it was beneficial to check if these interaction terms would help to create a more accurate model, we ultimately found that there were more useful techniques that would lead to better predictions.

In addition to the other nonlinear models we considered, we also revisited the randomForest package to predict *OREB* as we did with *SPREAD* and *TOTAL*. Like in our previous findings, the Random Forest model exhibited a slightly higher MAE of 4.163 compared to other methodologies, such as the linear models and log transformation model. Once again, this discrepancy appears rooted in overfitting during the model training, leading to an excessive inclusion of noise and subsequently skewing test results. While the Random Forest model can provide valuable insights and be a worthy alternative to linear models in intricate datasets, its overly complex nature consistently produced below average accuracy results compared to traditional linear models and generalized linear models.

An important property of offensive rebounds is that they are typically considered count data, representing the count of offensive rebounds made by a team in game. Following that property, the *OREB* variable in our dataset is the sum of the count of the home team's offensive rebounds and the away team's offensive rebounds.

Deviating from the linear regression models that we had been using, we decided to try implementing the use of Poisson regression to determine if we could find a better model that could predict the *OREB* values because this type of regression analysis is often used when the dependent variable represents counts and follows a Poisson distribution. Additionally, count data often displays non-negative and right-skewed distributions. To verify this, we plotted a histogram of the *OREB* variable and confirmed that its distribution was right skewed, indicating a potential beneficial use of Poisson regression.

We first took a logical approach in building these models, making very simple models that used a small number of predictors that we believed were influential in predicting the number of offensive rebounds. We attempted utilizing *FGA* and *FG3A*, as teams have more offensive rebounding opportunities when they miss more shots, and *FTA*, as offensive rebounds can occur when a team misses free throws because players can grab rebounds and extend possessions. We also used *REB*, as teams with a higher total rebound number might have more chances for offensive rebounds, and *AST*, because the number of assists might influence offensive rebounding by indicating ball movement and shots. We also used *TOV*, because as defensive rebounds often occur after turnovers, teams that commit more turnovers may give the opposing team more opportunities for offensive rebounds. Furthermore, we thought *PLUS_MINUS* could predict *OREB* values well because a positive value could correlate to more offensive rebounding

opportunities. We additionally hypothesized *FG_PCT* could be helpful because teams with lower field goal percentages might have more offensive rebounding opportunities because of more shots missed. Furthermore, other predictors in models that we used when determining whether a Poisson regression model would be beneficial included the *hrebounds_ratio*, as discussed in the Data Information section, representing the ratio of offensive rebounds and defensive rebounds, which we thought could be indicative of the rebounding dynamics of the game.

While we thought some variables could be helpful in predicting offensive rebounds, some variables in our data set did not seem relevant to the number of offensive rebounds such as *WL*, because whether the home or away team wins or loses is independent of total offensive rebounds. *MIN* might not be a strong predictor as most values are 48, and there was no significant difference between total rebounds in games that went to overtime and games that did. *FTM* is not related to misses or opportunities for offensive rebounds and as such does not have any explanatory power in predicting offensive rebounds. We also initially hypothesized that *STL* might not be a valuable predictor for offensive rebounds because the two statistics are independent, with there being no chance for an offensive rebound to occur after a steal since possession is lost. In addition, steals usually occur in-transition when players aren't close to the basket, which is primarily where offensive rebounds occur. *PF* is not a strong predictor for offensive rebounds because a personal foul is the end of a scoring opportunity for a team and there is no opportunity for an offensive rebound after a personal foul, except on the second free throw attempt once a team is in the bonus. Due to this ambiguity, *PF* has little predictive power in explaining offensive rebounds.

To compare the Poisson regression models we created, we initially thought to evaluate them based on AIC values, looking for lower values which would be indicative of a better model fit. Additionally, for each of the models we created using Poisson regression, we also created a linear model that used the same predictor variables so we could compare the MAE values between the two models to determine which model performed better based on a lower MAE value.

An initial simple model that was created to observe the performance of a Poisson regression model and to compare it to an identical linear model used *FGA* and *FG3A* for both home and away teams as predictor variables for *OREB*. While we initially knew interactions in this model would likely be beneficial to account for correlation and confounding between these predictor variables, for simplicity we will discuss the models without these terms in this paper.

The training data of the 2023 season was used as the data set for both models, and the family under the generalized linear regression model was specified as “Poisson.” The models can be shown below:

$$OREB_Poisson_Model1 = glm(OREB \sim HFGA + HF3GA + AFGA + AF3GA)$$

$$OREB_Linear_Model1 = lm(OREB \sim HFGA + HF3GA + AFGA + AF3GA)$$

The AIC value for the poisson model was calculated to be 5945.062, and the MAE value with the model's predictions for the training data was 17.701. This value was much higher than the simple linear regression's MAE value on the same data which was calculated as 4.341. This initial comparison between

models was surprising, as it was indicative that the Poisson regression models might perform worse than the simple linear regression models.

We continued making models with variables we believed to be meaningful and comparing between Poisson regression and simple linear regression models, adding more predictor variables to the models and utilizing various combinations of interaction terms to determine if AIC values could be lowered, but Poisson regression models consistently resulted in higher MAE values than simple linear regression models. One Poisson model we created that seemed to perform better than the other Poisson models due to a relatively lower AIC is displayed below:

$$OREB_Poisson_Model2 = glm(OREB \sim HTotalOREB + ATotalOREB + HFGA + AFGA + HBLK + ABLK + hrebound_pct + arebound_pct)$$

The model resulted in an AIC value of 5869.478 and an MAE value of 17.702. The Poisson model's corresponding simple linear regression model produced an MAE value of 4.211, thus making the Poisson model not as good as the simple linear model. However, an interesting finding from this model was that blocks seemed to have value in predicting offensive rebounds, possibly due to players repositioning themselves near the basket or disruptions that lead to missed shots and ultimately more rebound opportunities.

In a final attempt to observe if a Poisson regression performed more poorly than simple linear regression, we created a model identical to our best linear model for predicting offensive rebounds we created from stepwise regression. The model was as follows:

$$OREBModel1_Poisson = glm(OREB \sim HTotalOREB + ATotalOREB + HFG_PCT + AFG_PCT + HFGA + ASpread)$$

The AIC value of the model was 5810.469, lower than the other Poisson regression models we discussed in this paper and thus making it our best Poisson regression model, but the model resulted in a MAE error value of 17.704 compared to the simple linear regression model's MAE of 4.126 when fitting the data to the training set. Fitting the OREBModel1_Poisson model to the 2023-24 season, again using data from October 24, 2023, up until April 7, 2024, resulted in a MAE of 18.099.

From our attempts to find the best model to predict *OREB*, we came to the conclusion that using a Poisson regression model would likely not result in the most accurate predictions. Upon research to uncover potential explanations, we first found that the differences between a Poisson model and ordinary linear regression are that the Poisson model assumes errors that follow a Poisson distribution - not a normal distribution, and it models the natural log of the response variable as a linear function of the regression coefficients rather than modeling the response variable as a linear function of the regression coefficients. The Poisson model also assumes the mean is equal to the variance. In our training data set, the mean *OREB* value is 20.729 whereas the variance is 33.669. Because the variance was larger, over-dispersion could have been a large enough issue and a possible reason the Poisson model was not appropriate.

The following table summarizes the MAE of the actual and predicted *OREB* values during the 2023-24 season so far from top models that we tested:

Model	MAE for 2023-2024 <i>OREB</i>
Advanced Box Score Data	3.832
Advanced Box Score, Advanced Statistics, and Lineup Data	3.832
Log Transformation Model	3.851
Interaction Model	33.755
Random Forest Model	4.163
Poisson Model	18.099

The model that we used to make our prediction is as follows:

$$OREB = 1.023(HTotalOREB) + 1.279(ATotalOREB) + -2.032(HFG_PCT) + -0.121(AFG_PCT) + 0.002(HFGA) + 0.004(ASpread) - 21.313$$

To make our predictions on the last week of the 2024 season, we used the average advanced box score data from the 2023-24 season through April 7th, 2024. Since the games from April 9th to 14th do not have any data to predict from yet, we filled in the rows for each game with the home and away teams' averages for the part of the season that has happened. We did this by splitting the averaged data into a home and away dataset and then combining it with the predictions dataset we were given through a left join, similar to how we added the team advanced statistics and lineup data to our original cleaned dataset.